

Machine Learning Mock Exam: Modules 1 and 2

30 choice items built from the Module 1 and Module 2 slides. Answer key at the end.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Multiple Choice

1. A company wants a program that improves at spotting spam as it sees more emails. Which E, T, P part is the spam spotting?
 - A. The performance measure P
 - B. The training time
 - C. The experience E
 - D. The task T
2. Which statement contrasts rule-based programming and machine learning correctly?
 - A. Rule-based uses data; machine learning uses conditions
 - B. Both use only conditions
 - C. Rule-based uses conditions; machine learning uses data
 - D. Both use only labels
3. Why can machines suit routine, repetitive tasks better than people, per the slides?
 - A. They need no maintenance at all
 - B. They always cost more than humans
 - C. They work 24/7 and are not affected by fatigue or inattention
 - D. They never need any data
4. In the learning phase of the ML workflow, which step comes right after the training data?
 - A. Monitoring
 - B. Deployment
 - C. Preprocessing (clean and format the data)
 - D. Prediction

5. A dataset has blank cells, wrong records, and conflicting records. Which problems is it showing?
- A. Variance and bias
 - B. Overfitting and underfitting
 - C. Duplication only
 - D. Incompleteness, noise, and inconsistency
6. A boxplot is used to spot points far from the rest of the data. Which EDA activity is this?
- A. Outlier detection
 - B. Hypothesis testing
 - C. Feature scaling
 - D. Data reduction
7. After a model is deployed, its performance slowly worsens as real-world data changes. Which stage should catch this?
- A. Preprocessing
 - B. Monitoring
 - C. Validation
 - D. Training
8. A study causes unintended harm to a group even though the intentions were good. Which ethics principle is this about?
- A. Privacy
 - B. Ownership
 - C. Intention
 - D. Outcomes (disparate impact)
9. Predicting a house price is which kind of supervised task?
- A. Classification, because prices are labels
 - B. Reinforcement learning
 - C. Regression (the target is continuous)
 - D. Clustering

10. Filtering e-mails into spam or not spam is which kind of task?
- A. Binary classification
 - B. Clustering
 - C. Multiple classification
 - D. Regression
11. A model scores 99 percent on training data and 62 percent on test data. What is the likely problem?
- A. A scaling error only
 - B. Good generalization
 - C. Overfitting
 - D. Underfitting
12. A model scores 55 percent on both training and test data. What is the likely problem?
- A. Overfitting
 - B. Underfitting
 - C. Good generalization
 - D. Too much data
13. As model complexity keeps increasing, what happens to the training score and to generalization?
- A. Training drops, generalization improves
 - B. Both always improve
 - C. Neither changes
 - D. Training keeps improving, but generalization eventually drops
14. Which describes the best model on the complexity curve?
- A. The model with 100 percent training accuracy
 - B. The simplest model
 - C. The sweet spot between underfitting and overfitting
 - D. The most complex model

15. A rule is 100 percent accurate on only 12 rows. Why shouldn't you trust it yet?
- A. Twelve rows is the ideal size
 - B. Accuracy cannot be measured on training data
 - C. Many other rules would fit 12 rows equally well
 - D. Rules never work on small data
16. Which is the slide definition of bias?
- A. Error from sensitivity to noise
 - B. Error from wrong or too-simple assumptions in the learning algorithm
 - C. The difference between train and test sets
 - D. The size of the training set
17. In k-NN, increasing k from 1 to a large value moves the model toward what?
- A. Underfitting
 - B. Perfect memorization
 - C. A decision tree
 - D. Overfitting
18. On the plot of k-NN accuracy against n_neighbors, where is the best performance?
- A. At the largest n_neighbors
 - B. Somewhere in the middle
 - C. It is the same everywhere
 - D. At n_neighbors = 1
19. Which is a strength of k-NN listed in the slides?
- A. Great on sparse datasets
 - B. Fast with very many samples
 - C. Easy to understand and works well without special adjustments
 - D. Needs no distance measure
20. A linear regression on 105 features shows a huge gap between train and test R2. What should you try?
- A. Ridge regression to control complexity
 - B. Lower k in k-NN
 - C. Remove all regularization
 - D. Use fewer training rows

21. Which statement about Ridge vs Lasso is correct?

- A. Lasso can zero out features, while Ridge only shrinks them toward zero
- B. Both set every weight to exactly zero
- C. Ridge can zero out features, Lasso cannot
- D. Neither uses a penalty

22. Only a few of many features matter and you want an easy-to-read model. Which is the better pick per the slides?

- A. k-NN
- B. Lasso
- C. Ordinary least squares
- D. Ridge

23. Lasso with alpha 1.0 used only 4 of 105 features and scored badly. What fixes this?

- A. Removing the test set
- B. Raising alpha
- C. Lowering alpha
- D. Adding more duplicates

24. Which is true of LogisticRegression?

- A. It is a regression algorithm
- B. It cannot output probabilities
- C. It is a clustering method
- D. It is a classification algorithm despite its name

25. Why are Naive Bayes models efficient?

- A. They store the full dataset
- B. They search all feature interactions
- C. They build hundreds of trees
- D. They look at each feature individually and collect simple per-class statistics

26. A decision tree regressor is asked to predict outside the range of its training data. What happens?
- A. It always predicts zero
 - B. It extrapolates perfectly
 - C. It switches to a linear model
 - D. It cannot extrapolate beyond the training range
27. Which is a tree-based method that builds many independent trees on bootstrap samples?
- A. Ridge regression
 - B. Random forest
 - C. Gradient boosting
 - D. Kernel SVM
28. Gradient boosting hits 100 percent training accuracy. What are two ways the slides give to reduce overfitting?
- A. Remove the test set or add noise
 - B. Use more trees and no limit
 - C. Raise the learning rate or add depth
 - D. Limit the maximum depth or lower the learning rate
29. In an RBF-kernel SVM, a small gamma means what?
- A. A large kernel radius, so many points count as close
 - B. A linear kernel
 - C. A tiny radius
 - D. No regularization
30. Why must features be rescaled before an RBF SVM on the Breast Cancer data?
- A. SVMs only accept integer data
 - B. Scaling is required for k-NN only
 - C. Scaling removes the support vectors
 - D. The features have very different ranges and SVMs are sensitive to scale

Answer Key

No.	Answer	Why
1	D. The task T	M1: Formal definition
2	C. Rule-based uses conditions; machine learning uses data	M1: ML vs rule-based
3	C. They work 24/7 and are not affected by fatigue or inattention	M1: Why Machine Learning
4	C. Preprocessing (clean and format the data)	M1: How does ML work
5	D. Incompleteness, noise, and inconsistency	M1: Data Preprocessing
6	A. Outlier detection	M1: EDA
7	B. Monitoring	M1: Model Deployment
8	D. Outcomes (disparate impact)	M1: Data Ethics
9	C. Regression (the target is continuous)	M2: Supervised Learning
10	A. Binary classification	M2: Classification
11	C. Overfitting	M2: Overfitting
12	B. Underfitting	M2: Underfitting
13	D. Training keeps improving, but generalization eventually drops	M2: Model Complexity Curve
14	C. The sweet spot between underfitting and overfitting	M2: Model Complexity Curve
15	C. Many other rules would fit 12 rows equally well	M2: The boat example
16	B. Error from wrong or too-simple assumptions in the learning algorithm	M2: What is a Good Model
17	A. Underfitting	M2: Model complexity of k-NN
18	B. Somewhere in the middle	M2: Model Complexity of K-NN
19	C. Easy to understand and works well without special adjustments	M2: k-NN strengths
20	A. Ridge regression to control complexity	M2: Ridge Regression
21	A. Lasso can zero out features, while Ridge only shrinks them toward zero	M2: Ridge vs Lasso
22	B. Lasso	M2: Ridge vs Lasso
23	C. Lowering alpha	M2: Lasso
24	D. It is a classification algorithm despite its name	M2: LogisticRegression vs LinearSVC
25	D. They look at each feature individually and collect simple per-class statistics	Formative Q15
26	D. It cannot extrapolate beyond the	M2: Decision Tree Regression

	training range	
27	B. Random forest	M2: Random Forests
28	D. Limit the maximum depth or lower the learning rate	M2: Gradient Boosting Classifier
29	A. A large kernel radius, so many points count as close	M2: Tuning SVM Parameters
30	D. The features have very different ranges and SVMs are sensitive to scale	M2: Preprocessing Data for SVMs